What is PostgreSQL?

PostgreSQL is an free open-source database system that supports both relational (SQL) and non-relational (JSON) queries.

PostgreSQL is a back-end database for dynamic websites and web applications.

PostgreSQL supports the most important programming languages:

- Python
- Java
- C/C++
- C#
- Node.js
- Go
- Ruby
- Perl
- Tcl

Connect to the Database

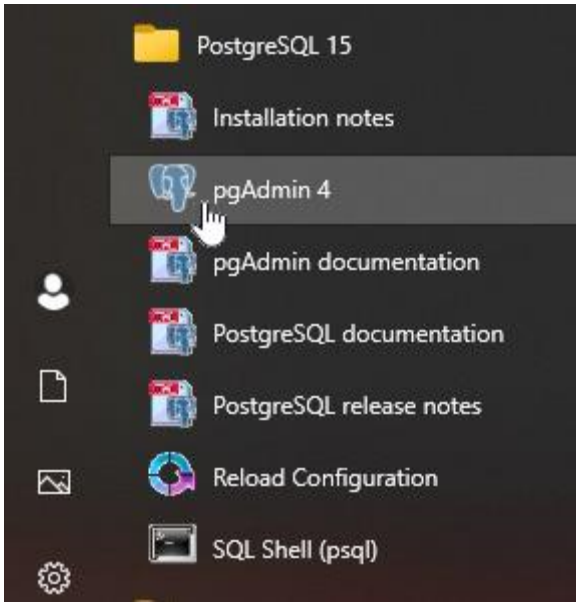
In the previous chapter we learned how to connect to the database using the SQL Shell (psql) application, which is a command-based application that allows us to interact with the PostgreSQL database.

There is another application that comes built-in with the PostgreSQL installation, the pgAdmin 4 application, which also offers to interact with the database, but in a more user-friendly way.

Let us take a look and see how it works.

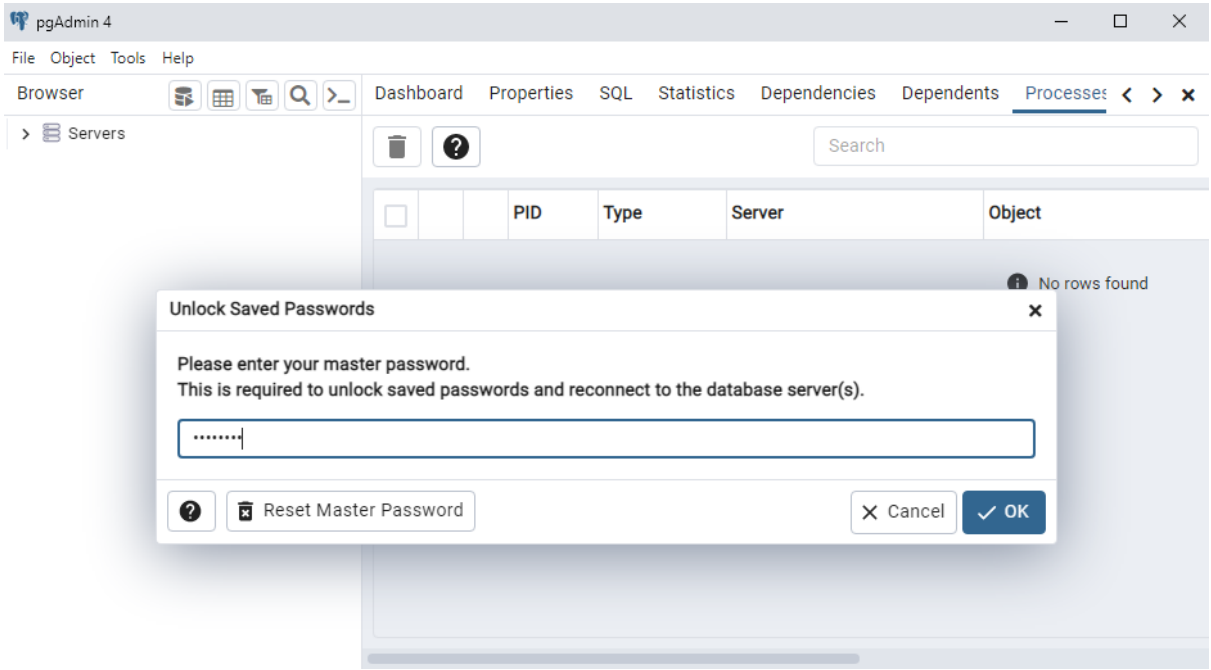
Start pgAdmin4

You will find the pgAdmin4 application in the start menu under PostgreSQL:



Tip: If you cannot find it, try searching for "pgAdmin4" on your computer.

Once the program has started, you should see a window like the one below, choose a master password, Since this is a local database that will run only on my computer, I will once again choose the password 12345678:



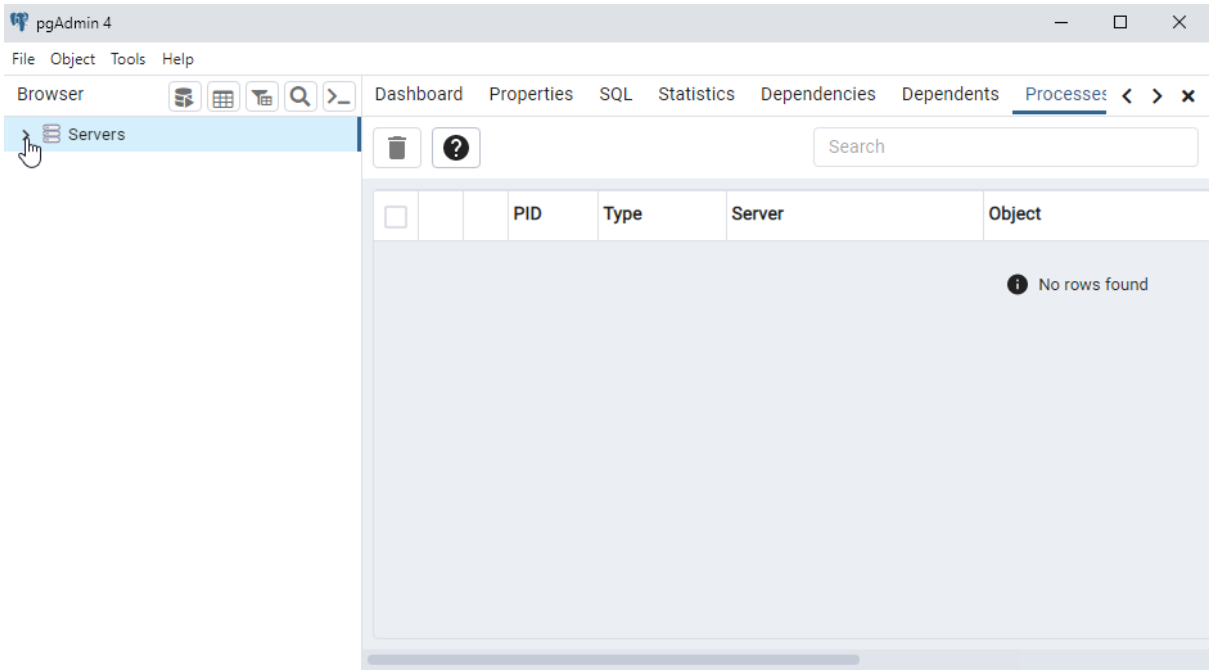
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pgAdmin4

Once you are inside the program, try to perform a simple SQL query.

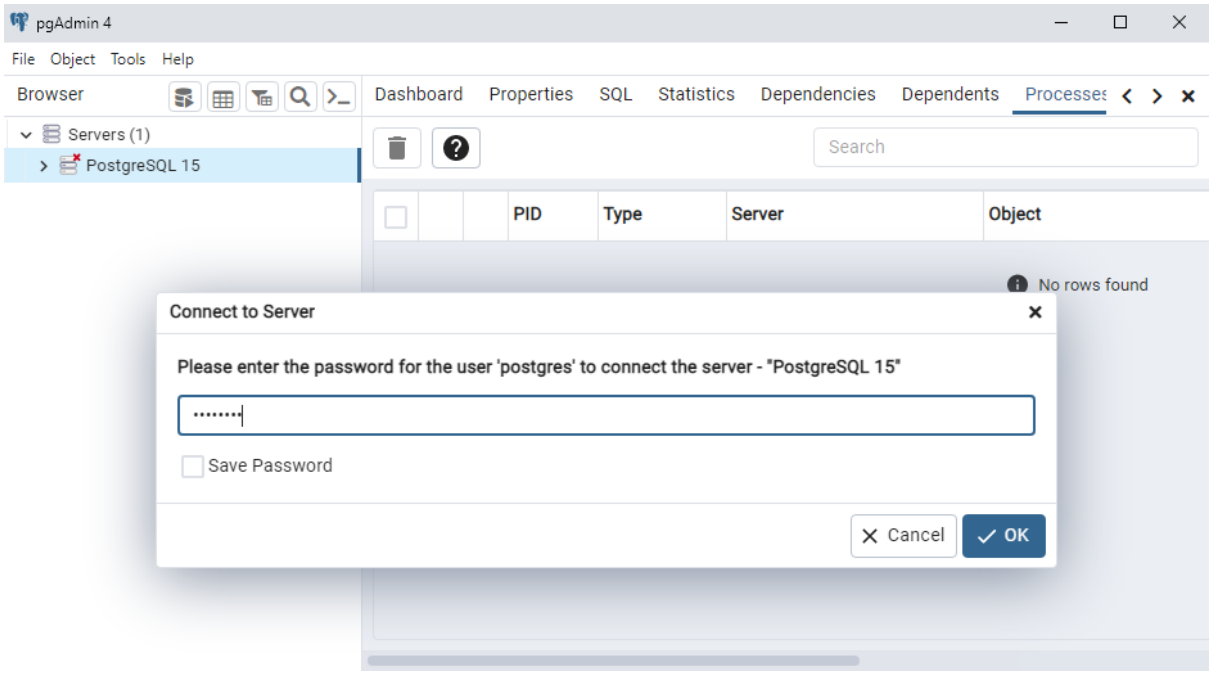
To do that we have to navigate to the database.

Start by opening the [Servers] option in the menu on the left:



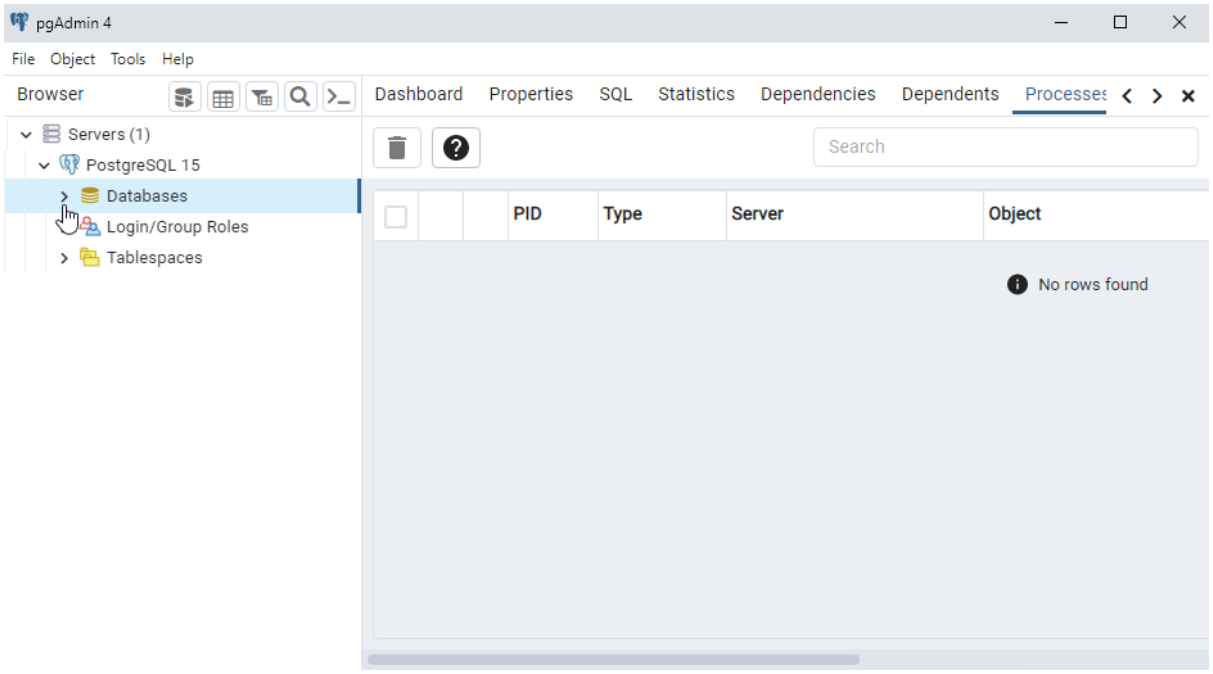
Connect to Server

Now you need to enter the password that you created when you installed PostgreSQL, my password is 12345678:



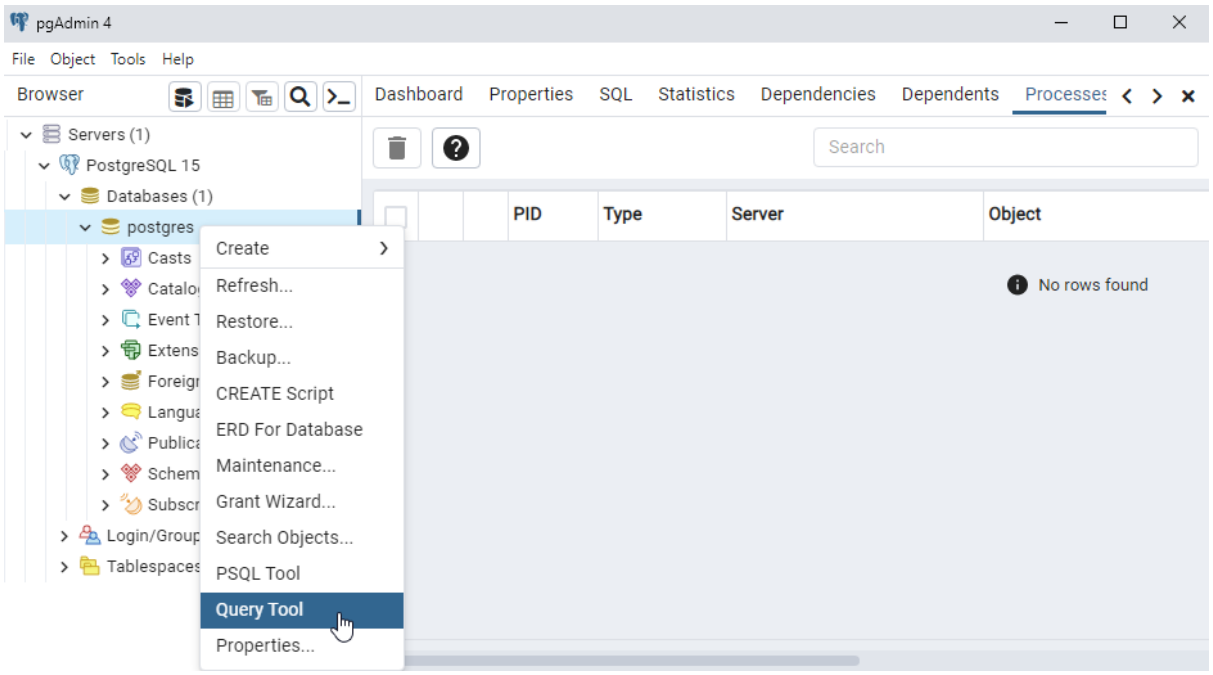
Find Database

Click on the [Database] option on in the menu on the left:



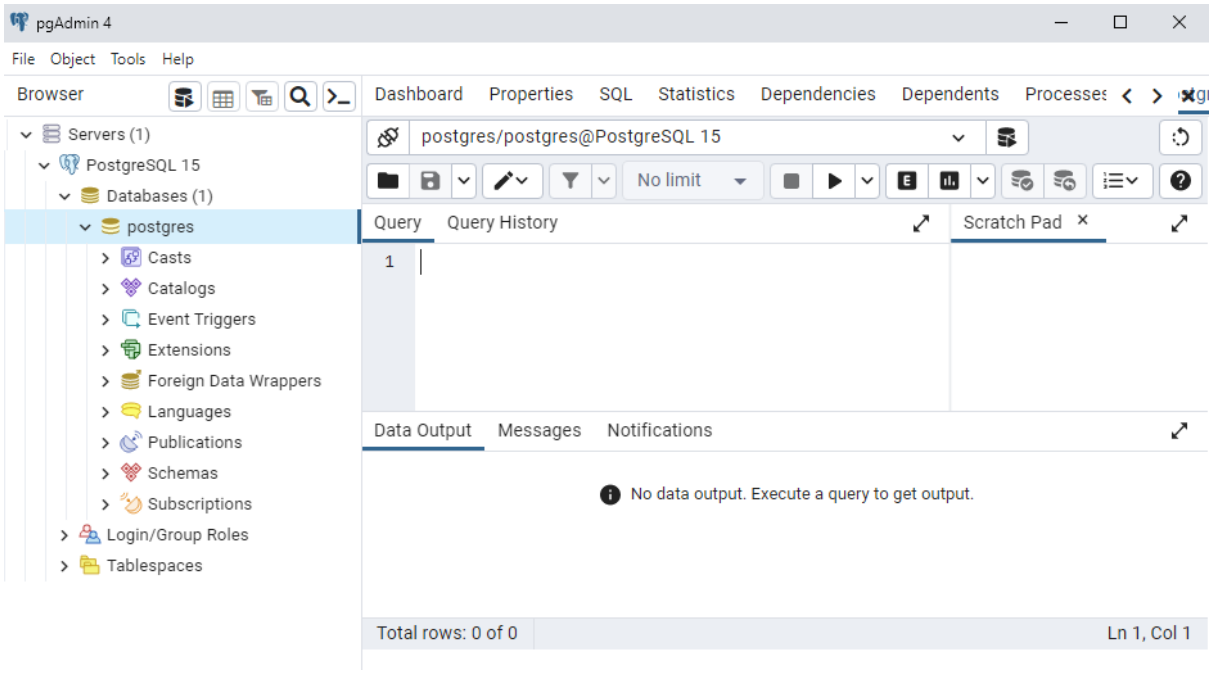
Open Query Tool

You should find a database named postgres, right-click it choose the "Query Tool":



Query Tool

In the Query Tool we can start executing SQL statements.

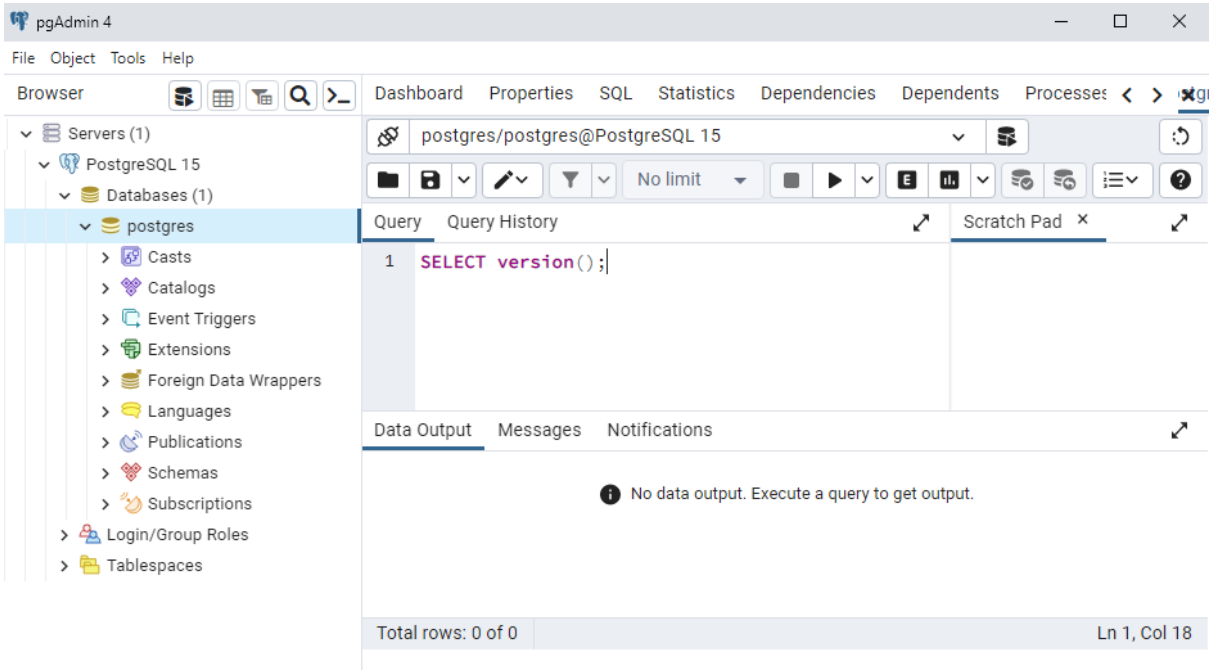


Write SQL Statements

Our database is empty, so we cannot query any tables yet, but we can check the version with this SQL statement:

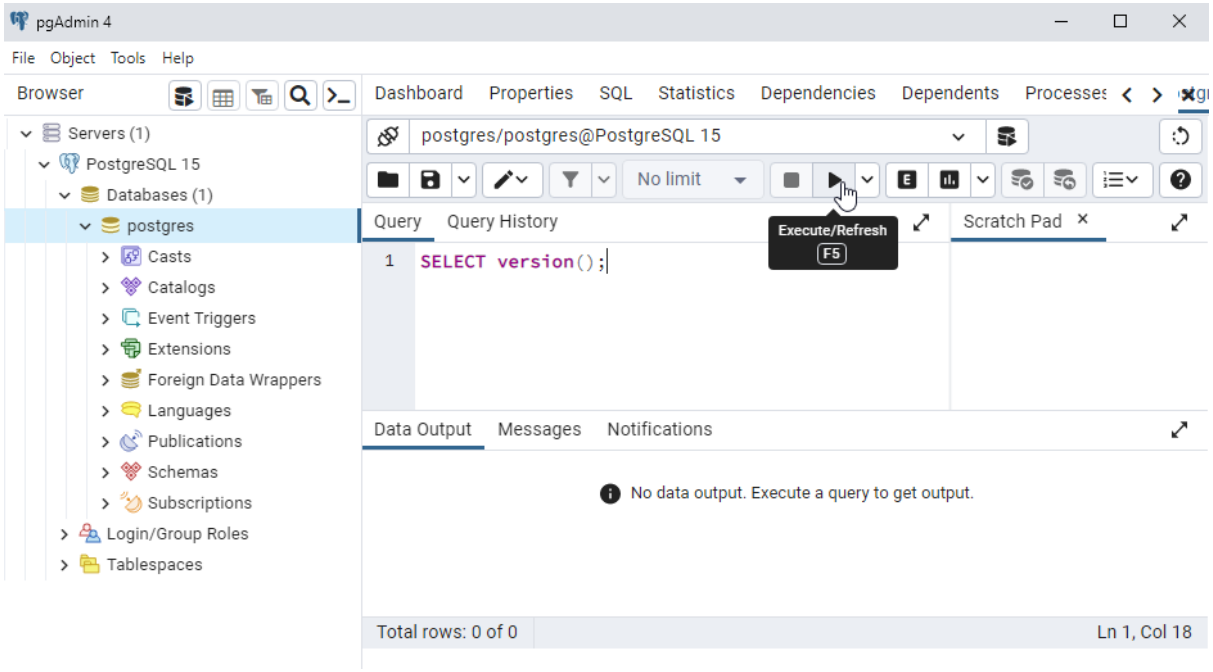
SELECT version();

To insert SQL statements in the Query Tool, just write in the input box like this:



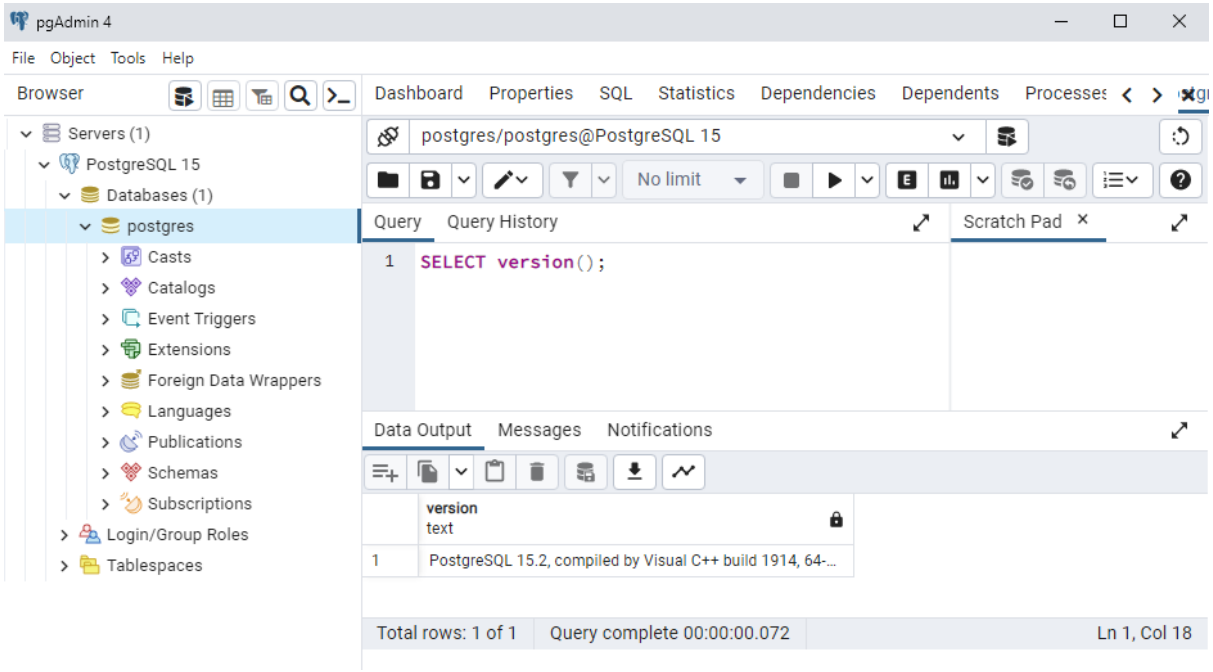
Execute SQL Statements

To execute a SQL statement, click the "Play" button above the input box:



Result

The SQL statement is executed, and you can see the result in the "Data Output" area:



Now we have learned two ways of connection to a database and execute SQL statements on it:

- SQL Shell (psql)
- pgAdmin 4

In the next chapters we will use the SQL Shell application to create tables and insert data into the database.

```
CREATE TABLE cars (
  brand VARCHAR(255),
  model VARCHAR(255),
  year INT
);

SELECT * FROM cars;


INSERT INTO cars (brand, model, year)
VALUES ('Ford', 'Mustang', 1964);

SELECT * FROM cars;

Insert Multiple Rows

INSERT INTO cars (brand, model, year)

VALUES

  ('Volvo', 'p1800', 1968),

  ('BMW', 'M1', 1978),

  ('Toyota', 'Celica', 1975);

SELECT * FROM cars;


Fetch the data

SELECT brand, year FROM cars;

SELECT * FROM cars;
```

ADD COLUMN

We want to add a column named color to our cars table.

```
ALTER TABLE cars
ADD color VARCHAR(255);

SELECT * FROM cars;
```

UPDATE Statement

```
UPDATE cars
SET color = 'red'
WHERE brand = 'Volvo';

SELECT * FROM cars;

UPDATE cars
SET color = 'red';

SELECT * FROM cars;

UPDATE cars
SET color = 'white', year = 1970
WHERE brand = 'Toyota';

SELECT * FROM cars;
```

The ALTER TABLE Statement

ALTER COLUMN

```
ALTER TABLE cars
ALTER COLUMN year TYPE VARCHAR(4);

ALTER TABLE cars
ALTER COLUMN color TYPE VARCHAR(30);
```

DROP COLUMN

```
ALTER TABLE cars
DROP COLUMN color;

SELECT * FROM cars;
```

The DELETE Statement

```
DELETE FROM cars
WHERE brand = 'Volvo';

SELECT * FROM cars;

DELETE FROM cars;

SELECT * FROM cars;
```

TRUNCATE TABLE

Because we omit the WHERE clause in the DELETE statement above, all records will be deleted from the cars table.

The same would have been achieved by using the TRUNCATE TABLE statement:

Example

Delete all records in the cars table:

```
TRUNCATE TABLE cars;
```

Result

```
TRUNCATE TABLE
```

Display Table

To check the result we can display the table with this SQL statement:

Example

```
SELECT * FROM cars;
```

The DROP TABLE Statement

```
DROP TABLE cars;
SELECT * FROM cars;
```